JOR Bayesian Fusion Framework

(For use with jor_fusion.py)

Purpose

This guide provides standardized instructions for entering inputs into the JOR Bayesian Fusion script. Following these procedures ensures consistent scoring, reproducibility, and defensible results.

1. Initial Prompts

Adjust Global Constants: Select **No** unless performing calibration.

- Default PRIOR_NH = 0.20
- Default CALIBRATION_K = 0.20

Evidence Weights: Ensure weights sum to 1.0.

- **C (Credibility) = 0.4**
- **E (Environment) = 0.3**
- **P (Physical) = 0.3**

Display Charts: Yes displays results immediately; **No** saves image files to the directory.

2. Case Identification

Enter a clear and descriptive case name for the database.
Examples: USS Nimitz 2004; Phoenix Lights; Airline Crew Sighting 2021.

3. Manual Entry Option

Use **only** if SOP (Solid Object Probability) and NHP (Non-Human Probability) values are already known.

- Enter values between 0.0 and 1.0.
- Otherwise, proceed with the full scoring breakdown below.

4. Scoring Categories

Witness Credibility (C)

Measures reliability and independence of observers.

- **Weak (0.30–0.50):** Single civilian, anonymous, no corroboration.
- **Moderate (0.55–0.65):** Two to three civilians or one trained observer.
- **Strong (0.70–0.80):** Multiple trained observers or independent witnesses.
- **Very Strong (0.81–0.85):** Trained personnel with independent documentation.

Modifiers: Each modifier is applied interactively via yes/no prompts:

- +0.03 logs
- +0.02 independent positions
- -0.03 inconsistencies
- -0.05 misidentification risk

Hard Caps: Optional; applied interactively via prompts:

- Single civilian max 0.50
- No trained observer max 0.70
- Anonymous max 0.45

Environmental / Observation Conditions (E)

Evaluates clarity and quality of observation conditions.

- **Weak (0.30–0.45):** Fog, heavy cloud, nighttime obscuration, brief duration.
- **Moderate (0.50–0.60):** Partial obstruction or moderate duration.
- **Strong (0.65–0.85):** Clear sky, controlled environment, extended duration.

Modifiers: Applied via prompts:

- +0.03 multiple vantage points
- +0.02 documented weather
- -0.03 distance >1 km
- -0.05 under 5 seconds

Hard Caps: Optional via prompts:

- Heavy fog max 0.40
- Nighttime single perspective max 0.70

Physical / Sensor Evidence (P)

Measures objective instrumentation and trace evidence.

- **Weak (0.30–0.45):** Anecdotal report only; no trace evidence.
- **Moderate (0.50–0.65):** One sensor or weak trace evidence.
- **Strong (0.70–0.85):** Two sensor types or confirmed anomalies.
- **Very Strong (0.86–0.95):** Multi-sensor confirmation or physical interaction.

Modifiers: Applied via prompts:

- +0.05 interference effects
- +0.03 long imagery
- +0.02 lab analysis
- -0.05 poor video
- -0.07 inconsistent sensors

Hard Caps: Optional via prompts:

- No sensor data max 0.55
- Video only max 0.75
- Multi-sensor max 0.95

5. Flight Behavior Classification

Adjusts anomaly weighting based on observed kinematics.

- **None / Conventional Flight (0.00):** Consistent with known aircraft or natural phenomena.
- **Minor Anomaly (+0.02):** Slightly unusual but explainable.

- **Moderate Anomaly (+0.04):** Clearly abnormal movement or trajectory.
- **Major Anomaly (+0.05):** Extreme maneuvers inconsistent with conventional aerodynamics.

Note: Flight behavior modifies the P component when computing NHP. SOP and NHP are calculated as **weighted sums** of C, E, and P (with flight adjustment applied to P).

Additional Note: All scored cases are logged automatically to jor_scores.csv.